

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester-I Examination, 2022-23

Botany

**Paper: IMB-101 — Microbial Physiology, Biochemistry, Genetics and
Molecular Biology**

Time: Three Hours

Maximum Marks: 70

Instructions: Answer five questions in all including, question 1 which is compulsory. The figures in the right hand margin indicate marks.

Question 1. Define or answer the following (not more than 50 words each): $[1 \times 10 = 10]$

- (a) Name any symbiotic cyanobacteria.
- (b) What is meant by generation time?
- (c) How many CO₂ molecules are produced per molecule of pyruvate fed to the TCA cycle?
- (d) What is the product of RNA polymerase II in eukaryotes?
- (e) What is Z-DNA?
- (f) What is vital mutation?
- (g) Who and when discovered the first transposable element in maize?
- (h) What is Shine-Dalgarno sequence?
- (i) Who discovered the ribozyme?
- (j) What is Michaelis-Menten constant (K_m)?

Question 2. Describe the structure of nitrogenase enzyme and the nitrogen fixation mechanism. $[15]$

Question 3. Write short notes on any two of the following: $[7.5 \times 2 = 15]$

- (a) Photophosphorylation in cyanobacteria
- (b) Entner-Doudoroff pathway
- (c) Sulphur cycle

Question 4. What is protein? Describe the α -helix and β -pleated sheets secondary structures of the protein. $[15]$

Question 5. Write short notes on any two of the following: $[7.5 \times 2 = 15]$

- (a) Substrate saturation curve of the enzyme

- (b) Types of lipids
- (c) Starch and glycogen

Question 6. Describe the DNA replication in detail with the help of suitable diagrams. [15]

Question 7. Write short notes on any two of the following: [7.5 × 2 = 15]

- (a) ρ -dependent and ρ -independent termination
- (b) Overlapping genes
- (c) Lac operon

Question 8. Briefly describe any two of the following: [7.5 × 2 = 15]

- (a) Transposons
- (b) Steps of the protein synthesis
- (c) Wobble hypothesis

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